UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/522,775	11/29/2023	Seong Jin CHOI	013057.2557	2695

89980 7590 04/01/2026

NSIP LAW
P.O. Box 65745
Washington, DC 20035

EXAMINER
VU, PHU

ART UNIT	PAPER NUMBER
2871	

NOTIFICATION DATE	DELIVERY MODE
04/01/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eofficeaction@appcoll.com
pto.nsip@gmail.com
pto@nsiplaw.com

Office Action Summary	Application No. 18/522,775	Applicant(s) CHOI et al.	
	Examiner PHU VU	Art Unit 2871	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 3/22/26.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Arguments

Applicant's arguments with respect to claim(s) 1-19 have been considered but are moot because the new ground of rejection does not rely on any reference applied in the prior rejection of record for any teaching or matter specifically challenged in the argument.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claim(s) 1-8 is/are rejected under 35 U.S.C. 103 as being unpatentable over Chen TWI7745252 in view of Miniutti US 2004/0145802 and further in view of JP6071143 (JP143)

Regarding claim 1, Chen teaches a lens assembly (fig. 1C display device 100 (includes the lens assembly)), comprising: a lens comprising an optical portion (first optical element 120) that refracts light and a flange portion (fig. 1C barrel portion 110) that extends from the optical portion; an optical film (linear/ reflective polarizer 2nd last paragraph of page 2) that contacts a first surface of the lens (curved lens see 2nd last paragraph of page 2); and a lens barrel (focusing barrel 140) configured to accommodate the lens and the optical film, wherein the optical film comprises a straight portion (L1 first cut edge) that is formed on a portion of a circumference of the optical film, and wherein the lens

comprises a linear portion (parallel to L1 edge) that is formed on the first surface of the lens, and is configured to contact the straight portion of the optical film.

Chen does not teach the optical film is fixed to the lens through an adhesive however Miniutti teaches optical films can easily be fixed through an adhesive [0046] to add optical effects compensation polarization etc.. Therefore, it would have been obvious to an ordinarily skilled artisan before the effective filing date of the claimed invention to fix the lens with the optical film through an adhesive to add optical effects compensation polarization etc.

Chen also does not teach the optical film comprises a V shaped inclined surface. However JP143 teaches forming notch marks (fig. 4 element 14) on a perimeter of optical films for alignment and reference purposes (page 2 last paragraph). While JP143 does not explicitly reference a V shaped inclined surface MPEP *(2144.04 section IV B - In re Dailey, 357 F.2d 669, 149 USPQ 47 (CCPA 1966) (The court held that the configuration of the claimed disposable plastic nursing container was a matter of choice which a person of ordinary skill in the art would have found obvious absent persuasive evidence that the particular configuration of the claimed container was significant.*) states changes this change in shape would considered obvious. Therefore, it would have been obvious to an ordinarily skilled artisan before the effective filing date of the claimed invention to modify Chen in view of JP143 to provide alignment and a reference point.

Regarding claim 2, Chen teaches the lens assembly of claim 1, wherein the linear portion of the lens (fig. 2A first cut edge L1) is formed inside an edge of the flange portion (110) of the lens.

Regarding claim 3, Chen teaches the lens assembly of claim 1, wherein the circumference of the optical film comprises the straight portion (L1) and an arc surface (see fig. 1C and 2A) connected to a first end of the straight portion and a second end of the straight portion.

Regarding claim 4, Chen teaches the lens assembly of claim 1, wherein the optical film comprises a polarizing film (linear polarizer see 2nd last paragraph of page 3) and a wave plate (page 5 3rd paragraph) that are stacked in an optical axis direction.

Regarding claim 5, Chen teaches the lens assembly of claim 4, wherein each of the polarizing film and the wave plate comprises the straight portion (first cut edge L1/ second cut edge L2).

Regarding claim 6, Chen teaches the lens assembly of claim 4 but does not explicitly teach wherein the optical film further comprises an anti-reflection film that is stacked on the polarizing film, and the anti-reflection film includes the straight portion.

While there is no direct recitation of an anti-reflection film one ordinary skill in the art could easily apply such a film in a similar manner as they are commonly associated with lens elements. When applied to a lens similar to Hirata this would already have first and second linear and straight portions enabling good alignment and optical effect (see first two paragraphs after disclosure heading). Therefore, it would have been obvious to one of ordinary skill in the art to modify Hirata in view of Chen to enable good alignment and optical effect.

Regarding claim 7, Chen teaches the lens assembly of claim 1, wherein a coating layer (semi-reflection coating see page 5 3rd paragraph) is formed on at least one of the first surface of the lens and a second surface of the lens.

Regarding claim 8, Chen teaches the lens assembly of claim 1, wherein the lens further comprises a decut (fig. 2A 110 outer straight portion parallel to L1) portion that is formed on an edge of the flange portion of the lens.

Claim(s) 9-19 is/are rejected under 35 U.S.C. 103 as being unpatentable over Hirata JP2006-201378 in view of Chen TWI7745252 in view of JP6071143 (JP143) and further in view of Miniutti US 2004/0145802.

Regarding claim 9, Hirata teaches a lens assembly, comprising: a lens comprising an optical portion (fig. 7-8 lens 2) that refracts light and a flange portion (barrel 1) that extends from the optical portion; wherein the lens comprises a first linear portion (contact section 32a) and a second linear portion (flat surface portion 12).

Hirata does not teach an optical film that contacts a first surface of the lens; and a lens barrel configured to accommodate the lens and the optical film, wherein the optical film comprises a first straight portion that extends in a first direction from an arbitrary first outer point of the optical film, and a second straight portion that extends in a second direction from the arbitrary first outer point of the optical film, and a first linear portion that contacts the first straight portion and a second linear portion that contacts the second straight portion.

However Chen teaches an optical film (polarizer page 5 paragraph 2) that contacts a first surface of the lens; and a lens barrel (140) configured to accommodate the lens and the optical film wherein the optical film follows the shape of the lens (page 5 paragraph 2). This combination would render obvious the limitations of the optical film comprises a first straight portion that extends in a first direction from an arbitrary first outer point of the optical film, and a second straight portion that extends in a second direction from the arbitrary first outer point of the optical film, and a first linear portion that contacts the first straight portion and a second linear portion that contacts the second straight portion as one of ordinary skill in the art could easily apply polarizer films and or wave plates with straight edge that follows the shape of the lens (page 5 paragraph 2-3). When applied to a lens similar to Hirata this would already have first and second linear and straight portions enabling good alignment and optical effect (see first two paragraphs after disclosure heading). Therefore, it would have been obvious to one of ordinary skill in the art to modify Hirata in view of Chen to enable good alignment and optical effect.

Hirata as modified by Chen teach the lens assembly of claim 9, wherein each of the first linear portion of the lens (Hirata fig. 7 and 8 firm contact sections 32a) and the second linear portion (gate trace section 13) of the lens is formed inside an edge of the flange portion (barrel 1) of the lens.

Hirata and Chen does not teach the optical film is fixed to the lens through an adhesive however Miniutti teaches optical films can easily be fixed through an adhesive [0046] to add optical effects compensation polarization etc.. Therefore, it would have been obvious to an ordinarily skilled artisan before the effective filing date of the claimed invention to fix the lens with the optical film through an adhesive to add optical effects compensation polarization etc.

Regarding claim 11, Hirata as modifed by Chen teach th ens assembly of claim 9, wherein the first straight portion of the optical film (would follow contact section 32a) and the second straight portion of the optical film (would flat surface portion 12) form an obtuse angle with each other.

Regarding claim 12, Hirata as modifed by Chen teaches the lens assembly of claim 9, wherein a distance from the first outer point of the optical film to an optical axis in a direction perpendicular to the optical axis is less than a radius of the optical film (flat surface portion 12 is somewhat recessed vs the radius of the optical film).

Regarding claim 13, Chen teaches optical film comprises a polarizing film and a wave plate that are stacked in an optical axis direction (page 5 paragraph 2-3).

Regarding claim 14, the combination of Hirata as modified by Chen would teach the lens assembly of claim 13, wherein each of the polarizing film and the wave plate comprises the first straight portion (32) and the second straight portion (12).

Regarding claim 15, Chen teaches the lens assembly of claim 9, wherein a coating layer can be formed on at least one of the first surface of the lens and a second surface of the lens (page 5 paragraph 3).

Regarding claim 16, Chen teaches the lens further comprises a decut portion (fig. 2a outer surface of barrel 110) that is formed on an edge of the flange portion of the lens.

Regarding claim 17, Hirata teaches a lens assembly, comprising: a lens comprising an optical portion (fig. 7-8 lens 2) and a flange portion (barrel 1); the first linear portion (contact portion 32a) of the lens and a second linear portion (flat portion 12) of the lens

Hirata does not teach an optical film that contacts a first surface of the lens; wherein the lens comprises a first linear portion and a second linear portion, wherein the optical film comprises an anti-reflection film, a polarizing film, and a wave plate, wherein each of the anti-reflection film, the polarizing film, and the wave plate of the optical film comprises a first straight portion and a second straight portion, and wherein the first linear portion of the lens is aligned with the first straight portion of the optical film, and the second linear portion of the lens is aligned with the second straight portion of the optical film.

Chen teaches an optical film that contacts a first surface of the lens; wherein the lens comprises a first linear portion and a second linear portion, wherein the optical film comprises an anti-reflection film, a polarizing film, and a wave plate, wherein each of the anti-reflection film, the polarizing film, and the wave plate of the optical film comprises a first straight portion and a second straight portion, and wherein the first linear portion of the lens is aligned with the first straight portion of the optical film, and the second linear portion of the lens is aligned with the second straight portion of the optical film.

However Chen teaches an optical film (polarizer page 5 paragraph 2) that contacts a first surface of the lens; and a lens barrel (140) configured to accommodate the lens and the optical film wherein the optical film follows the shape of the lens (page 5 paragraph 2) these optical films being polarizers and/or waveplates to the lenses (page 5 paragraph 2-3)

This combination would render obvious the limitations wherein the lens comprises a first linear portion and a second linear portion, wherein the optical film comprises, a polarizing film, and a wave

plate, wherein each of the polarizing film, and the wave plate of the optical film comprises a first straight portion and a second straight portion, and wherein the first linear portion of the lens is aligned with the first straight portion of the optical film, and the second linear portion of the lens is aligned with the second straight portion of the optical film. While there is no direct recitation of an anti-reflection film one ordinary skill in the art could easily apply such a film in a similar manner as they are commonly associated with lens elements. When applied to a lens similar to Hirata this would already have first and second linear and straight portions enabling good alignment and optical effect (see first two paragraphs after disclosure heading). Therefore, it would have been obvious to one of ordinary skill in the art to modify Hirata in view of Chen to enable good alignment and optical effect.

Chen does not teach the optical film is fixed to the lens through an adhesive however Miniutti teaches optical films can easily be fixed through an adhesive [0046] to add optical effects compensation polarization etc.. Therefore, it would have been obvious to an ordinarily skilled artisan before the effective filing date of the claimed invention to fix the lens with the optical film through an adhesive to add optical effects compensation polarization etc.

Regarding claim 18, Chen teaches a single straight portion whose ends are connected through an arc surface of the film (see fig. 2C) and Hirata teaches a sequence of straight portions which extend to form a quasi arc surface of the film (see fig. 7 and 8). While neither teach exactly two joined by an arc surface this is considered a matter of obvious design choice as the current application also presents one straight portion and does not meaningfully distinguish between one or two straight portions. Given the teaches of Chen (one straight surface) and Hirata (a sequence of straight surfaces) this is considered an obvious variant.

Regarding claim 19, Chen teaches the optical film is fixed to the first surface of the lens page 5 paragraph 2.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to PHU VU whose telephone number is (571)272-1562. The examiner can normally be reached 11:00 - 7:00 M-F.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jennifer Carruth can be reached at 571-272-9791. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/PHU VU/
Primary Examiner, Art Unit 2871